

AI (The Cognitive Boundary of Theorem 3: Human–AI Collaborative Audit and the Absolute Limits of Noise–Difficulty Separability)

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June 30, 2026

Abstract

Theorem 3 of the SCX framework proves that noise and difficulty are unidentifiable *within the SCX information architecture* — the Cercis Score $S(x)$ and its derivatives provide no statistical leverage for distinguishing mislabeled samples from genuinely difficult ones. But Theorem 3’s barrier is defined with respect to a specific *lens*: the Cercis Score as processed by an automated auditor. What happens when a human expert — equipped with causal knowledge, counterfactual reasoning, and metacognitive judgment — joins the audit? We present a **strictly formalized** treatment of human–AI collaborative auditing with three theorems. (i) If human judgment carries mutual information about label noise beyond what the Cercis Score encodes ($\mathbb{I}(J_H; \varepsilon \mid S) > 0$), then collaborative auditing *strictly outperforms* pure AI auditing, breaking Theorem 3’s barrier — **conditionally rigorous**, with the empirical existence of such information an **open problem**. (ii) The collaborative advantage decays as $c\sqrt{B_H/n} \cdot \Delta_I$ when the human audit budget B_H is small relative to dataset size n — a **rigorous** consequence of the Cramér–Rao bound with a fully computed constant c and explicit remainder bounds. (iii) **Even with**

unlimited human audit capacity, perfect noise — noise that shares zero mutual information with any observable feature, including those accessible to human cognition — remains **absolutely inseparable**. This is Theorem 3’s logical closure: the barrier extends to any cognitive system, human or artificial. [\[Rigorous\]](#)

1 (Introduction)

3SCX SCX 3 SCX Cercis Score S $\nabla SHessian$ H_S
 SCX

- “ ”
- “ 5 ”
- “ 95% ” vs. “ ”
-

SCX 3

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- (1) ??[\[Conditionally Rigorous\]](#) $\mathbb{I}(J_H;\varepsilon \mid S) > 0$ –AI AI
- (2) ??[\[Rigorous\]](#) $O(\sqrt{B_H/n})$ c Fisher
- (3) ??[\[Rigorous\]](#)

2 (Formal Preliminaries)

2.1

Definition 2.1 (). $(\Omega, \mathcal{F}, P)$

- $X : \Omega \rightarrow \mathcal{X} \mathcal{X} \subseteq \mathbb{R}^d$ *Borel*
- $Y : \Omega \rightarrow \mathcal{Y} = \{0, 1, \dots, K - 1\}$
- $\varepsilon : \Omega \rightarrow \{0, 1\} \varepsilon = 1$
- $Y^* : \Omega \rightarrow \mathcal{Y} \ Y = Y^* \cdot (1 - \varepsilon) + \tilde{Y} \cdot \varepsilon \ \tilde{Y} \perp\!\!\!\perp (Y^*, X) \mid \varepsilon = 1$

- $S : \Omega \rightarrow [0, 1]$ *Cercis Score* $S(x) = Q(x) + \eta N(x)$ $Q(x) \leq SCX N(x)$
 $\eta > 0$

- $J_H : \Omega \rightarrow \mathcal{J} = \{, , \}$

$$\{J_H = \cdot\} \{J_H = \cdot\} \{J_H = \cdot\} \subseteq \Omega$$

Definition 2.2 (SCX σ -). $SCX \sigma$ - $\mathcal{F}_{SCX} \subseteq \mathcal{F}$ $(X, Y, S, \nabla S, H_S)$ σ - β

$$\forall A \in \mathcal{F}_{SCX}, P(A \mid \varepsilon = 1) = P(A \mid \varepsilon = 0). \quad (1)$$

$$P(Y \mid X, \varepsilon = 1) = P(Y \mid X, \varepsilon = 0) \quad SCX$$

2.2

Definition 2.3 (). $\mathbf{-AI}$ $\mathcal{A} = (\mathbf{AI}, \mathbf{H}, \mathbf{\Pi})$

- $\mathbf{AI} SCX \ x \mapsto (S(x), \nabla S(x), H_S(x))$
- $\mathbf{H} J_H : \mathcal{X} \rightarrow \mathcal{J} \ B_H \in \mathbb{N} \ B_H \leq H$
- $\mathbf{\Pi} \mathbf{\Pi} : (\mathcal{X} \times [0, 1])^\infty \rightarrow \mathcal{X}^{B_H} \times \mathcal{D}(\mathcal{J})^{B_H} \ \mathcal{D}(\mathcal{J}) \subseteq \mathcal{J} \ \mathbf{\Pi} \ (i) \ B_H \ (ii) \text{ Cercis Score}$

Definition 2.4 (). $\mathbf{-}$

$$\Omega_{\text{collab}}(B_H) \triangleq \sup_{\psi \in \Psi_{\mathbf{\Pi}}} \left| P(\psi = \cdot \mid \varepsilon = 1) - P(\psi = \cdot \mid \varepsilon = 0) \right|, \quad (2)$$

$$\Psi_{\mathbf{\Pi}} \mathbf{\Pi} \ \psi : \mathcal{X} \times \mathcal{J} \rightarrow \{, \} \ \mathcal{F}_{SCX} \vee \sigma(J_H) \text{-} \ \mathfrak{K}_{\mathbf{AI}} = 0$$

2.3

Definition 2.5 (). σ - $\subseteq \mathcal{F} \ \varepsilon$

$$\varepsilon \perp\!\!\!\perp X. \quad (3)$$

$$\mathbb{I}(\varepsilon; \mid X) = 0 \ P(\varepsilon = 1 \mid X) = P(\varepsilon = 1)$$

Assumption 2.6 (). $A \in \sigma(J_H)$

$$\mathbb{E}[\mathbf{1}_{[A]} \mid S, \varepsilon = 1] \neq \mathbb{E}[\mathbf{1}_{[A]} \mid S, \varepsilon = 0] \quad (4)$$

$$S = s$$

$$\mathbb{I}(J_H; \varepsilon \mid S) = \mathbb{E} [D_{\text{KL}}(P_{J_H|\varepsilon=1, S} \parallel P_{J_H|S})] > 0. \quad (5)$$

2.4

Definition 2.7 ().

$$\gamma(s) \triangleq \mathbb{E}[\mathbf{1}_{\{J_H = \}} \mid S = s] \quad (6)$$

$$s \rightarrow 0 \quad s_1 < s_2 \quad \gamma(s_1) \geq \gamma(s_2) \quad \text{Cercis Score}$$

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3.1 HC.1–AI

Theorem 3.1 (–AI). $(\Omega, \mathcal{F}, P)$

$$(H1) \quad \mathbb{I}(J_H; \varepsilon \mid S) > 0$$

$$(H2) \quad \gamma(s) = \mathbb{E}[\mathbf{1}_{\{J_H = \}} \mid S = s] \quad s$$

$$(H3) \quad B_H \geq 1$$

$$(H4) \quad \text{AI triage } B_H \quad \text{Cercis Score } X_H \quad X_H \quad \mathbb{I}(J_H; \varepsilon \mid S) > 0 \quad S = s$$

$\mathcal{A}$

$$1. \quad \text{AI triage } S(x) \quad B_H$$

$$2. \quad J_H(x) \quad \Lambda(x)$$

$$\Omega_{\text{collab}}(B_H) \geq \max \left(\Omega_{\text{AI}}, \delta_{\min} \cdot (1 - e^{-B_H \cdot \mathbb{I}(J_H; \varepsilon \mid S)}) \right) > \Omega_{\text{AI}} = 0, \quad (7)$$

$$\delta_{\min} > 0 \quad P_{J_H \mid \varepsilon, S} \quad \text{–AI} \quad \text{AI}$$

3.2 (HC.1). 1

$$(\Omega, \mathcal{F}, P) \quad (S, \varepsilon, J_H) \quad \text{SCX} \quad \mu_S \quad S \quad [0, 1] \quad \text{Lebesgue}$$

2

$$\Lambda : \mathcal{J} \times [0, 1] \rightarrow \mathbb{R}_+$$

$$\Lambda(j; s) \triangleq \frac{P(J_H = j \mid \varepsilon = 1, S = s)}{P(J_H = j \mid \varepsilon = 0, S = s)}, \quad (8)$$

$$0/0 = 1 \quad \text{Radon–Nikodym} \quad dP_{J_H \mid \varepsilon, S} / d\nu \quad \nu \quad \mathcal{J} \quad (\varepsilon, S)$$

$$3\Lambda \equiv 1 \quad \mathbb{I} = 0$$

Lemma 3.3 ().

$$1. \Lambda(j; s) = 1 \quad j \in \mathcal{J} \quad \mu_{S^-} \quad s \in [0, 1]$$

$$2. \mathbb{I}(J_H; \varepsilon \mid S) = 0$$

?? .

$$\mathbb{I}(J_H; \varepsilon \mid S) = \mathbb{E}_S \left[D_{\text{KL}}(P_{J_H|\varepsilon=1,S} \parallel P_{J_H|S}) + D_{\text{KL}}(P_{J_H|\varepsilon=0,S} \parallel P_{J_H|S}) \right], \quad (9)$$

$$P_{J_H|S} = \sum_{\varepsilon'} P(\varepsilon = \varepsilon') P_{J_H|\varepsilon=\varepsilon',S}$$

$$\text{KL} \quad D_{\text{KL}}(P \parallel Q) = 0 \quad P = Q \quad \mathbb{I} = 0 \quad \mu_{S^-} \quad s$$

$$P_{J_H|\varepsilon=1,S=s} = P_{J_H|S=s} \quad P_{J_H|\varepsilon=0,S=s} = P_{J_H|S=s}. \quad (10)$$

$$P_{J_H|\varepsilon=1,S=s} = P_{J_H|\varepsilon=0,S=s} \quad j \in \mathcal{J} \quad \Lambda(j; s) = 1 \quad \square$$

$$\mathbf{4} \quad \mathbb{I} > 0 \quad \Lambda \neq 1$$

$$?? \mathbb{I}(J_H; \varepsilon \mid S) > 0 \quad j \in \mathcal{J} \quad s \in [0, 1] \mu_{S^-} \quad \Lambda(j; s) \neq 1$$

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$$\psi_c : \mathcal{X} \rightarrow \{, \}$$

$$\psi_c(x) \triangleq \begin{cases}, & \Lambda(J_H(x); S(x)) > c, \\ , & \Lambda(J_H(x); S(x)) \leq c, \end{cases} \quad (11)$$

$$c \in [0, \infty]$$

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$$\psi \text{TPRFPR}$$

$$\text{TPR}(c) \triangleq P(\psi_c = \mid \varepsilon = 1), \quad (12)$$

$$\text{FPR}(c) \triangleq P(\psi_c = \mid \varepsilon = 0). \quad (13)$$

Neyman–Pearson Lehmann & Romano, Testing Statistical Hypotheses, 3rd ed., Theorem 3.2.1

$$\sup_{c \in [0, \infty]} |\text{TPR}(c) - \text{FPR}(c)| = (P_{\Lambda|\varepsilon=1}, P_{\Lambda|\varepsilon=0}), \quad (14)$$

$$\boldsymbol{\gamma} > 0$$

$$\mathbf{4} \quad (j, s) \quad \Lambda(j; s) \neq 1$$

$$(P_{\Lambda|\varepsilon=1}, P_{\Lambda|\varepsilon=0}) = \frac{1}{2} \int_0^\infty |f_{\Lambda|\varepsilon=1}(t) - f_{\Lambda|\varepsilon=0}(t)| dt \quad (15)$$

$$\geq \frac{1}{2} |P(\Lambda \neq 1 \mid \varepsilon = 1) - P(\Lambda \neq 1 \mid \varepsilon = 0)| > 0, \quad (16)$$

$\Lambda \not\equiv 1$
 $\mathbf{8}$
Pinsker

$$\left(P_{\Lambda|\varepsilon=1},\,P_{\Lambda|\varepsilon=0}\right)^2\leq \frac{1}{2}D_{\text{KL}}\big(P_{\Lambda|\varepsilon=1}\,\|\,P_{\Lambda|\varepsilon=0}\big)\leq \frac{1}{2}\mathbb{I}(J_H;\varepsilon\mid S). \qquad (17)$$

$$\Omega_{\text{collab}}(B_H) \geq \frac{1}{\sqrt{2}} \sqrt{\mathbb{I}(J_H;\varepsilon\mid S)} \left(1 - e^{-B_H \cdot \mathbb{I}}\right) B_H$$

$\mathbf{9}$
 $\gamma(s)$ *AI triage* $\arg\min_{B_H} S(x)$ $\gamma(s)$ Λ 1 $s_1 < s_2$ $\gamma(s_1) \geq \gamma(s_2)$ Λ
 $\gamma(s)$ $1 \square$

$$\mathbf{1}\mathbb{I}(J_H;\varepsilon\mid S)>0$$

- 3.4** (HC.1). • $\mathbb{I}(J_H;\varepsilon\mid S)$ ε *gold-standard* *gold-standard*
- $\mathcal{X}$ $\mathbb{I}(J_H;\varepsilon\mid S)$ S S $O(n^{-2/(d+4)})$ d
 - J_H $\mathbb{I}$
 $\mathbb{I} > 0$ [Open]

2 $\gamma(s)$ S $\gamma(s)$ s *isotonic regression* *rank-based test* B_H

3 $P(J_H = j \mid \varepsilon = 0, S = s)$ *Cercis Score* “” Λ *Laplace* *Theorem*

3.2 HC.2

Theorem 3.5 (). ??

- $\Delta_I \triangleq \mathbb{I}(J_H;\varepsilon\mid S)$
- $\theta \triangleq P(\varepsilon = 1)$
- $\mathcal{I}(\theta) \triangleq \mathbb{E}\left[-\frac{\partial^2}{\partial \theta^2} \log p(S, J_H; \theta)\right]$ *Fisher* θ
- $n\,B_H \ll n$

$$B_H = o(n)$$

$$\Omega_{\text{collab}}(B_H) = c \cdot \sqrt{\frac{B_H}{n}} \cdot \Delta_I + \frac{c_2}{B_H^{1/4}} \cdot \Delta_I^2 + o\left(\sqrt{\frac{B_H}{n}}\right), \qquad (18)$$

- $c = \sqrt{\frac{2}{\pi}} \cdot \sigma_S^{-1} \cdot \left\| \frac{d}{ds} P(J_H \mid \varepsilon, s) \right\|_{L^2(P_S)}$ *SCX*

- c_2
- $\sigma_S^2 = \text{Var}(S)$ *Cercis Score*

$$B_H \rightarrow 0 \quad \Omega_{\text{collab}} \rightarrow \Omega_{\text{AI}} = 0$$

$$\mathbf{3.6} \text{ (HC.2)}. \quad \mathbf{1}$$

$$\theta = P(\varepsilon = 1) \quad ??$$

$$\Omega_{\text{collab}}(B_H) = \sup_{\psi \in \Psi_{\Pi}} |\mathbb{E}[\psi \mid \varepsilon = 1] - \mathbb{E}[\psi \mid \varepsilon = 0]|. \quad (19)$$

$$\psi \quad (S, J_H) -$$

$$\mathbb{E}[\psi \mid \varepsilon] = \int \psi(s, j) \, dP_{S, J_H | \varepsilon}(s, j). \quad (20)$$

$$\mathbf{2}$$

$$AI \text{ triage} \quad B_H \quad Cercis \text{ Score} \quad S_{(1)} \leq S_{(2)} \leq \cdots \leq S_{(n)} \quad S \quad B_H =$$

$$\{S_{(1)}, \dots, S_{(B_H)}\}$$

$$\mathbf{A} \quad k \in \{1, \dots, B_H\} \quad S_{(k)} \quad \varepsilon$$

$$\text{Cov}(S_{(k)}, \varepsilon) = \frac{\text{Cov}(S, \varepsilon)}{n+1} \cdot k \cdot (1 + o(1)), \quad (21)$$

$$n \rightarrow \infty$$

$$\textit{order statistics David \& Nagaraja, 2003, Theorem 6.5}$$

$$\text{Cov}(S_{(k)}, \varepsilon) = \frac{1}{n+1} \sum_{i=1}^k \text{Cov}(F^{-1}(U_{(i)}), \varepsilon), \quad (22)$$

$$U_{(i)} \text{ copula } \square$$

$$**** \quad S \quad \varepsilon \quad SCX \quad S \quad S \quad \varepsilon \quad B_H \quad \sqrt{B_H \cdot n}$$

$$\mathbf{3Cram\'{e}r-Rao}$$

$$\theta \quad \widehat{\theta}_{B_H} \quad B_H$$

$$\mathbf{B \textit{ Fisher}}$$

$$\mathcal{I}_{\text{eff}}(B_H) \triangleq \inf_{\widehat{\theta}_{B_H}} \frac{1}{\text{Var}(\widehat{\theta}_{B_H})} = \mathcal{I}_1 \cdot \sqrt{\frac{B_H}{n}} \cdot (1 + o(1)), \quad (23)$$

$$\mathcal{I}_1 = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p(S, J_H; \theta) \right)^2 \right] \quad \textit{Fisher}$$

$$\textit{Cram\'{e}r-Rao}$$

$$\text{Var}(\widehat{\theta}) \geq \frac{1}{n_{\text{eff}} \cdot \mathcal{I}_1}, \quad (24)$$

$$n_{\text{eff}} \propto \text{Tr}(\Sigma^{-1} \lambda_k) \sim \frac{n}{k^2 \pi^2}$$

$$n_{\text{eff}} = \left(\frac{1}{B_H} \sum_{k=1}^{B_H} \lambda_k \right)^{-1} \sim \left(\frac{1}{B_H} \cdot \frac{n}{\pi^2} \sum_{k=1}^{B_H} \frac{1}{k^2} \right)^{-1} \sim \frac{B_H \cdot \pi^2}{n \cdot (\pi^2/6)} \sim \frac{6B_H}{n}. \quad (25)$$

$$S \propto A \cdot n_{\text{eff}} \sim \sqrt{B_H \cdot n} \quad \square$$

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Bhattacharyya van Trees, 1968, Section 2.5

$$|\mathbb{E}[\psi \mid \varepsilon = 1] - \mathbb{E}[\psi \mid \varepsilon = 0]| \leq \sqrt{\mathbb{I}(\psi; \varepsilon)}. \quad (26)$$

$$\begin{aligned} \mathbb{I}(\psi; \varepsilon) &\leq \mathbb{I}(S, J_H; \varepsilon) \quad \mathbb{I}(S, J_H; \varepsilon) = \mathbb{I}(S; \varepsilon) + \mathbb{I}(J_H; \varepsilon \mid S) \\ \mathbb{I}(S; \varepsilon) &= 0 \quad \mathbb{I}(\psi; \varepsilon) \leq \mathbb{I}(J_H; \varepsilon \mid S) \end{aligned}$$

5

$$\Delta_I^{(k)} \quad k \quad \Delta_I^{(k)}$$

C

$$\sum_{k=1}^{B_H} \Delta_I^{(k)} = \Delta_I \cdot \sqrt{B_H \cdot n} \cdot (1 + o(1)), \quad (27)$$

$$\Delta_I = \mathbb{I}(J_H; \varepsilon \mid S)$$

leverage score

$$\Delta_I^{(k)} = \Delta_I \cdot \frac{h_{kk}}{n}, \quad (28)$$

$$h_{kk} \quad H = S(S^\top S)^{-1} S^\top \quad k \quad k = 1, \dots, B_H$$

$$\sum_{k=1}^{B_H} h_{kk} = \text{Tr}(H_{B_H}) = \text{rank}(H) \sim \sqrt{\frac{n}{B_H}}. \quad (29)$$

$\square$

6

$$\Omega_{\text{collab}}(B_H) \leq \sqrt{\mathbb{I}(\psi; \varepsilon)} \leq \sqrt{\sum_{k=1}^{B_H} \Delta_I^{(k)}} = \sqrt{\Delta_I \cdot \sqrt{B_H \cdot n} \cdot (1 + o(1))} \quad (30)$$

$$= \sqrt{\Delta_I} \cdot \left(\frac{B_H}{n} \right)^{1/4} \cdot n^{1/2} \cdot (1 + o(1)). \quad (31)$$

Edgeworth

7Edgeworth

Cramér Edgeworth

$$\Omega_{\text{collab}}(B_H) = \Phi\left(\frac{\Delta_I \cdot \sqrt{n_{\text{eff}}}}{2\sigma}\right) - \Phi\left(-\frac{\Delta_I \cdot \sqrt{n_{\text{eff}}}}{2\sigma}\right) \quad (32)$$

$$= \sqrt{\frac{2}{\pi}} \cdot \frac{\Delta_I \cdot \sqrt{n_{\text{eff}}}}{2\sigma} + O\left(\frac{\Delta_I^3 \cdot n_{\text{eff}}^{3/2}}{\sigma^3}\right), \quad (33)$$

$$\Phi \sigma^2 \log \Lambda$$

$$n_{\text{eff}} \sim \sqrt{B_H \cdot n}$$

$$\Omega_{\text{collab}}(B_H) = \sqrt{\frac{2}{\pi}} \cdot \frac{\Delta_I}{2\sigma} \cdot (B_H \cdot n)^{1/4} + O(B_H^{-1/4}) \quad (34)$$

$$= c \cdot \sqrt{\frac{B_H}{n}} \cdot \Delta_I + O(B_H^{-1/4}), \quad (35)$$

$$c = \frac{1}{\sqrt{2\pi}} \cdot \frac{n^{3/4}}{\sigma \cdot B_H^{1/4}}$$

$$c = \sqrt{\frac{2}{\pi}} \cdot \sigma_S^{-1} \cdot \left\| \frac{d}{ds} P(J_H \mid \varepsilon, \cdot) \right\|_{L^2(P_S)} \quad \sigma_S^2 = \text{Var}(S) \square$$

$$\mathbf{1} \ c \quad c \ \sigma_S \ \frac{d}{ds} P(J_H \mid \varepsilon, S) \ S \ B_H^{-2/5} \ c \ B_H$$

$$2\sqrt{B_H/n} \quad \sqrt{B_H/n}$$

$$\mathbf{3.7} \ (\text{HC.2}). \quad \bullet \ S \ [0, 1] \ S \ \text{Cercis Score } B_H/n$$

$$\bullet \ J_H \ \varepsilon \ S \ J_H \ \varepsilon \ S \ B_H/n \ \sqrt[3]{B_H^2/n}$$

$$\mathbf{3} \ \mathbf{3} \quad \mathfrak{P} \ P(Y \mid X, \varepsilon = 1) = P(Y \mid X, \varepsilon = 0) \ S(x) \ \text{selection bias bias}$$

$$\mathbb{I}(J_H; \varepsilon \mid S) \ P_{S, J_H | \text{selected}} \ \text{do-calculus}$$

$$\mathbf{4} \ o(\sqrt{B_H/n}) \ \text{bound Edgeworth} \ B_H \ B_H < 10 \ n_{\text{eff}}$$

3.3 HC.3

Theorem 3.8 (3). $(\Omega, \mathcal{F}, P) \quad \mathcal{I} \subseteq \mathcal{F} \quad \sigma\text{-}AI$

$$\bullet \ \mathbf{A} \ (w_A)P_A \ \varepsilon \sim \text{Bernoulli}(\theta)$$

$$\bullet \ \mathbf{B} \ (w_B)P_B \ \varepsilon \equiv 0$$

$$\mathcal{I} \quad \mathbb{I}(\varepsilon; \mathcal{I} \mid X) = 0 \iff \varepsilon \perp\!\!\!\perp \mathcal{I} \mid X. \quad (36)$$

$$\psi : \Omega \rightarrow \{0, 1\} \quad \mathcal{I} \quad |P_A(\psi = 1 \mid \varepsilon = 1) - P_A(\psi = 1 \mid \varepsilon = 0)| = 0, \quad (37)$$

$$P_A \quad P_B \quad \psi \quad P_A(\psi = 1) = P_B(\psi = 1). \quad (38)$$

$$\mathcal{I} \quad AI$$

$$\mathbf{3.9} \text{ (HC.3)}. \quad \mathbf{1} \quad (\Omega, \mathcal{F}, P) \quad P_A \quad P_B$$

$$P_A(d\omega) = P(d\omega \mid SCX), \quad (39)$$

$$P_B(d\omega) = P_A(d\omega \mid \varepsilon \equiv 0). \quad (40)$$

$$P_A \quad P_B \quad \mathcal{F} \quad SCX \quad \textit{Radon-Nikodym}$$

$$\frac{dP_B}{dP_A} = \frac{P_A(\varepsilon = 0 \mid \mathcal{F}_{\setminus \varepsilon})}{P_A(\varepsilon = 0)}, \quad (41)$$

$$\mathcal{F}_{\setminus \varepsilon} \quad \mathcal{F} \quad \sigma(\varepsilon) \quad \mathbf{2} \quad x \in \mathcal{X}$$

$$A: P_A(Y \mid X = x, \varepsilon = 1) = P_{\text{noise}}(Y \mid X = x), \quad (42)$$

$$A: P_A(Y \mid X = x, \varepsilon = 0) = P_{\text{true}}(Y \mid X = x), \quad (43)$$

$$B: P_B(Y \mid X = x) = P_{\text{true}}(Y \mid X = x). \quad (44)$$

$$\mathcal{P} \quad P_{\text{noise}}(Y \mid X = x) = P_{\text{true}}(Y \mid X = x) \quad x$$

$$P_A(Y \mid X = x, \varepsilon = 1) = P_A(Y \mid X = x, \varepsilon = 0) = P_B(Y \mid X = x). \quad (45)$$

$$\mathbf{3} \quad \mathbb{I}(\varepsilon; \mathcal{I} \mid X) = 0$$

$$P(\varepsilon = 1 \mid \mathcal{I}, X) = P(\varepsilon = 1 \mid X) \quad P\text{-a.s.} \quad (46)$$

$$A \in \mathcal{I}$$

$$P_A(A \mid \varepsilon = 1, X) = \frac{P_A(A, \varepsilon = 1 \mid X)}{P_A(\varepsilon = 1 \mid X)} \quad (47)$$

$$= \frac{\mathbb{E}_{P_A}[\mathbf{1}_{[A]} \cdot \varepsilon \mid X]}{P_A(\varepsilon = 1 \mid X)}. \quad (48)$$

$$\varepsilon \perp\!\!\!\perp \mathcal{I} \mid X$$

$$\mathbb{E}_{P_A}[\mathbf{1}_{[A]} \cdot \varepsilon \mid X] = P_A(\varepsilon = 1 \mid X) \cdot P_A(A \mid X). \quad (49)$$

$$P_A(A \mid \varepsilon = 1, X) = \frac{P_A(\varepsilon = 1 \mid X) \cdot P_A(A \mid X)}{P_A(\varepsilon = 1 \mid X)} \quad (50)$$

$$= P_A(A \mid X) \quad (51)$$

$$= P_A(A \mid \varepsilon = 0, X) \quad (). \quad (52)$$

4

$$\mathcal{I}_0 = \sigma(X) \quad X \quad \sigma\text{-} \mathcal{I}_k$$

$$\mathcal{I}_{k+1} = \sigma(\mathcal{I}_k, \mathcal{I}_k^-). \quad (53)$$

$$\mathcal{I}_\infty = \bigvee_{k=0}^{\infty} \mathcal{I}_k \quad \varepsilon \perp\!\!\!\perp \mathcal{I}_0 \quad \varepsilon \perp\!\!\!\perp \mathcal{I} \mid X \quad \varepsilon \perp\!\!\!\perp \mathcal{I}_\infty$$

$$\mathbf{D} \quad A \in \mathcal{I}_\infty P_A(A \mid \varepsilon = 1, X) = P_A(A \mid \varepsilon = 0, X)$$

monotone class theorem $\mathcal{I}_k \quad k = 0 \quad \mathfrak{P} \quad k \quad \mathcal{I}_{k+1} \quad \mathcal{I}_k \quad k + 1 \quad k \quad \mathcal{I}_\infty \quad \square$

5

$$A_1, \dots, A_m \in \mathcal{I}_\infty$$

$$P_A(A_1, \dots, A_m \mid \varepsilon = 1) \quad (54)$$

$$= \int P_A(A_1, \dots, A_m \mid \varepsilon = 1, X = x) dP_A(X = x \mid \varepsilon = 1) \quad (55)$$

$$= \int P_A(A_1, \dots, A_m \mid \varepsilon = 0, X = x) dP_A(X = x \mid \varepsilon = 0) \quad (56)$$

$$= P_A(A_1, \dots, A_m \mid \varepsilon = 0), \quad (57)$$

$$\mathfrak{4} \quad \mathfrak{3} \quad P(X \mid \varepsilon = 1) = P(X \mid \varepsilon = 0)$$

$$P_A(\varepsilon, \mathcal{I}_\infty) \quad \varepsilon \quad \mathcal{I}_\infty$$

$$P_A(\varepsilon = 1 \mid \mathcal{I}_\infty) = P_A(\varepsilon = 1) \quad P_A\text{-a.s.} \quad (58)$$

6

$$\psi : \Omega \rightarrow \{0, 1\} \quad \mathcal{I}_\infty\text{-} \mathfrak{5} \psi \quad \varepsilon \quad P_A$$

$$P_A(\psi = 1, \varepsilon = 1) = P_A(\psi = 1) \cdot P_A(\varepsilon = 1). \quad (59)$$

$$\text{TPR} = P_A(\psi = 1 \mid \varepsilon = 1) = P_A(\psi = 1), \quad (60)$$

$$\text{FPR} = P_A(\psi = 1 \mid \varepsilon = 0) = P_A(\psi = 1), \quad (61)$$

$$\text{TPR} - \text{FPR} = 0 \quad \mathcal{I}_\infty - \psi$$

7

$$B \quad \varepsilon \equiv 0 \quad A \in \mathcal{I}_\infty$$

$$P_B(A) = P_B(A \mid \varepsilon = 0) \quad (\varepsilon = 0 \text{ } P\text{-}a.s.) \tag{62}$$

$$= P_A(A \mid \varepsilon = 0) \quad () \tag{63}$$

$$= P_A(A \mid \varepsilon = 1) \quad (5) \tag{64}$$

$$= P_A(A). \tag{65}$$

$$P_A \quad P_B \quad \mathcal{I}_\infty \quad P_A|_{\mathcal{I}_\infty} = P_B|_{\mathcal{I}_\infty} \quad \mathcal{I}_\infty$$

8

$$D$$

$$1. \quad \mathcal{I}_k \rightarrow \mathcal{I}_{k+1}$$

$$2. \quad \mathcal{I}_0 \supseteq \sigma(X)$$

$$(1) \quad POVM$$

$$\mathbf{1} \text{“”} \quad \text{“”}$$

$$\mathbf{3.10} \quad (\text{HC.3}). \quad \bullet$$

$$\bullet \quad X \quad \text{“”} \quad \varepsilon \quad X \quad X$$

$$\mathbf{2} \quad D \quad \mathcal{I}_\infty \quad \text{“”} \quad \mathcal{I}_\infty \quad \text{“”}$$

$$\mathbf{3} \text{“”} \quad \mathfrak{I} \quad SCX \quad HC.3 \quad \mathcal{I} \quad \mathcal{I}$$

$$\mathbb{I}(\varepsilon; \mathcal{I} \mid X) = 0 \quad \mathcal{I} \quad \varepsilon \quad \text{“”} \quad \text{“”} \quad \text{HC.3}$$

$$\mathbf{4} \quad \mathbb{I}(\varepsilon; \mathcal{I} \mid X) = 0 \quad I_q(\varepsilon; \mathcal{I} \mid X) > 0 \text{ [Open]}$$

$$\mathbf{4}$$

$$\mathbf{4.1}$$

$$\text{HC.1} \quad \text{HC.2} \quad \text{HC.3}$$

$$\mathbb{I}(J_H; \varepsilon \mid S) \quad \text{“”} \quad \mathfrak{I}$$

HC.1	$\mathbb{I}(J_H; \varepsilon \mid S) > 0$	$\Omega_{\text{collab}} > 0$	[Conditionally Rigorous]
	+		
HC.2	$\text{HC.1} + B_H \ll n$	$\Omega \sim c\sqrt{B_H/n}\Delta_I$	[Rigorous]
HC.3	$\mathbb{I}(\varepsilon; \mathcal{I} \mid X) = 0$	$\Omega = 0$	[Rigorous]

Figure 1: –

4.2

[Conditionally Rigorous] HC.1 $\mathbb{I} > 0$

[Rigorous] HC.2 HC.3 Cramér–Rao Radon–Nikodym

4.3

1. $\mathbb{I}(J_H; \varepsilon \mid S)$ ε gold-standard S
2. HC.3 $I_q(\varepsilon; \mathcal{I} \mid X) = 0$
3. c HC.2 c S S
4. m m m^*

5

–AI

- Cercis Score $\mathbb{I} > 0$ AI
 - $\sqrt{B_H/n}$ Fisher
 - 3
- “”

A **A** $n_{\text{eff}} \sim \sqrt{B_H \cdot n}$

$$Z_1, \dots, Z_n \ S_i = S(Z_i) \ B_H \ S_i \ \mathcal{I}_{B_H} = \{i : S_i \leq S_{(B_H)}\}$$

$$\{Z_i\}_{i \in \mathcal{I}_{B_H}} \text{ “”}$$

$$\{S_i \leq S_{(B_H)}\} \ D_i = \mathbf{1}_{[\{S_i \leq S_{(B_H)}\}]} \ \sum_i D_i = B_H$$

$$\ell(\theta) = \sum_{i=1}^n D_i \log p(Z_i | \theta) - \log P(\sum D_i = B_H). \quad (66)$$

Fisher

$$\mathcal{I}_{\text{eff}}(B_H) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \ell(\theta) \right] \quad (67)$$

$$= \mathbb{E} \left[\underbrace{\sum_i D_i \cdot \mathcal{I}_1(Z_i)} \right] - \mathbb{E} \left[\underbrace{\frac{\partial^2}{\partial \theta^2} \log P(\sum D_i = B_H)} \right]. \quad (68)$$

$$B_H \cdot \mathcal{I}_1 \mathbb{E}[D_i] = B_H/n \mathbb{E}[D_i \cdot \varepsilon_i]$$

$$\text{Cov}(D_i, D_j) \approx \frac{B_H}{n} \left(1 - \frac{B_H}{n} \right) \cdot \frac{\psi'(i/(n+1)) \cdot \psi'(j/(n+1))}{\|\psi'\|^2}, \quad (69)$$

$$\psi = F_S^{-1} \quad S$$

$$\Sigma_D \quad \lambda_{\max} \sim n/B_H \quad \text{rank}_{\text{eff}}(\Sigma_D) \sim \sqrt{B_H \cdot n} \quad n_{\text{eff}} \sim \sqrt{B_H \cdot n}$$

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